Kuku AI Generalist Assignment

Problem 1: Find the Promos (Ads) — Automated Engine & Strategy Show: "I Married a Broken Soldier. He was a Duke" (50 Episodes)

Candidate Submission

1. PROBLEM CHOICE & FRAMING

Why Problem 1: Vertical microdramas depend directly on paid user acquisition across TikTok, Reels, and YouTube Shorts. For a show to succeed, 90% of viewership funnels through high-CTR 20–60s promotional clips. Today, human editors manually scrub through 50–100 episodes per title, creating a severe operational bottleneck as Kuku releases new series weekly. Solving promo discovery unlocks scalable, instant marketing pipelines.

CORE ASSUMPTIONS (WHERE BRIEF WAS SILENT)

- **Strict Time Ceiling & Retention Sweet-Spot:** 20.0s is the absolute floor; 60.0s is the hard ceiling. Peak virality occurs in the **35s–50s** sweet spot (sufficient runway for setup, escalation, and cliffhanger without viewer drop-off).
- **Cold Viewer Comprehension:** Ads target non-subscribers. Candidates reliant on deep lore or secondary gossip are heavily discounted against universal, primal conflict archetypes.
- **Cross-Episode Multi-Segment Joins:** Permitted only when causal continuity is preserved (e.g., Ep 41 explosion immediately resolved by Ep 42 Duke entrance).
- **Acoustic Decay Buffer:** Written script timestamps clip abruptly; an automated **+0.40s** padding is required in video assembly to preserve lingering vocal reverb.

SCORING RUBRIC (0–100 TOTAL SCORE)

- **Hook Strength (30%):** Stop-scroll impact within 0–3.0s (slaps, weapons, physical fall, outrageous insults, high emotional intensity tags).
- **Cliffhanger Suspense (30%):** Unresolved tension at cut-point (sound crescendo, blade frozen, unanswered proposal/question, blackout).
- **Standalone Coherence (25%):** Comprehensibility to cold viewer without backstory (clear hero/villain dynamic, low lore penalty).
- **Pacing & Event Density (15%):** Penalizes dragging exposition; rewards punchy back-and-forth dialogue within optimal duration.

2. APPROACH & ARCHITECTURE: THE TWO-STAGE HYBRID ENGINE

Passing 120,000 words of raw scripts directly into an LLM causes severe timestamp hallucinations, boundary jitter, and astronomical API costs. Instead, we built a **Two-Stage Funnel Architecture** combining deterministic code for speed and structure, with semantic intelligence for evaluation:

`STAGE 1: CODE & RULES`

Parser & Window Generator

Parses 50 episodes into 3,727 structured events with millisecond precision. Generates valid 20–60s windows and cross-episode candidate segments in **0.75 seconds**.

`STAGE 2: SEMANTIC SCORER`

Multi-Factor Virality Engine

Evaluates 0–3s hook triggers, cut-point cliffhangers, and lore burden. Deduplicates overlapping segments using **Non-Maximum Suppression (IoU < 0.35)**.

`STAGE 3: HUMAN-IN-THE-LOOP`

Interactive Web Dashboard

Marketing editors review ranked cards, inspect synced script excerpts, view diagnostic quality flags, and export CSV/JSON timeline manifests.

UNIT ECONOMICS & PRODUCTION SCALE (100 SHOWS/MONTH)

Token Cost: Deterministic pre-filtering strips 85% of redundant text, sending only candidate windows (~18k tokens/show) to lightweight LLMs (Gemini 1.5 Flash / GPT-4o-mini). Total AI cost is **~$0.04 per 50-episode show** ($4.00/month for 100 shows) vs. $2,500/month for brute-force prompting. Total execution time is under 1 second per show.

3. STRATEGIC PROMO RANKING & MARKETING FUNNEL

Viral success requires matching promo archetypes to viewer lifecycle stages. We classify detected promos into a 3-tier conversion funnel:

Rank	Episode & Time	Dur.	Archetype & Phase	Hook (0–3s) & Cliffhanger Cut	Why It Works & Strategic Rationale
#1	**Ep 41 & 42** 41: 01:17.5–01:36.0 42: 00:00.0–00:23.4	41.9s	`SECRET DUKE REVEAL` **Phase 3: Paywall**	**Hook:** Cathedral entrance detonates in a massive fireball. **Cliff:** True Duke marches in smoke; Royal Guard refuses arrest: *"We will not."*	**Peak Wish-Fulfillment:** Hollywood-scale physical explosion coupled with instant power reversal. Post at paywall drop to drive coin purchases.
#2	**Ep 1 & 3** 1: 01:50.0–02:17.0 3: 01:58.0–02:25.7	54.8s	`WARRIOR PROPOSAL` **Phase 1: Acquisition**	**Hook:** Gerald rejects bride at altar for her sister. **Cliff:** Scarred warrior drops to one knee with ring: *"Will you marry me, Juliana?"*	**Outrage to Romance Rollercoaster:** Extreme moral injustice instantly redeemed by a mysterious protector. Maximum top-of-funnel TikTok CTR.
#3	**Ep 12** 12: 00:15.0–00:38.0 12: 00:58.0–01:27.2	52.2s	`BRIAR CRAWL TRIAL` **Phase 2: Binge Hook**	**Hook:** Clarissa: *"Crawl frozen blackthorn briars barehanded, or watch him hang!"* **Cliff:** Juliana reaches into razor thorns as Alexander screams.	**Visceral Empathy & High Stakes:** Sadistic impossible choice triggers intense emotional distress, driving viewers to install the app to see if she survives.
#4	**Episode 1** 01:26.0–02:18.0	52.0s	`ALTAR BETRAYAL` **Phase 1: Acquisition**	**Hook:** Juliana begs father to look at her; groom sneers. **Cliff:** Devastation turns to icy resolve as cello drone peaks into blackout.	**Pure Outrage Bait:** Zero background lore needed. Viewers immediately side with the betrayed bride and demand the arrogant fiancé's retribution.
#5	**Episode 32** 02:11.8–02:34.4	22.6s	`DUNGEON STANDOFF` **Phase 2: Binge Hook**	**Hook:** Chained Juliana: *"Look at you... you've already lost."* **Cliff:** Red-hot iron lowers; extreme close-up on unblinking blue eye.	**Intense Psychological Duel:** Unbroken defiance against brutal torture creates unbearable micro-suspense right before physical strike.

4. QUALITY CHECKING: WHERE DOES THE TOOL PICK BADLY?

In accordance with the evaluation brief, an honest tool must be transparent about its limits. Out of 148 detected candidates, our diagnostic engine categorized: **Tier S (90+): 5** | **Tier A (80-89): 7** | **Tier B (70-79): 31** | **Tier C (<70): 105**. We identified four distinct failure modes where automated heuristics fail without human intervention:

FAILURE MODE 1: THE FALSE ACTION HOOK (VISUAL VS. TEXT)

The Defect: Dialogue contains intense conflict keywords (e.g. *"I know what you did"*), but the camera direction indicates two characters seated drinking tea with zero motion.

Why It Fails: Keyword scanners award high hook points based on verbal confrontation, but TikTok scrollers swipe away in <1.2s without immediate visual action or rapid camera punch-in.

Fix Implemented: The parser classifies camera and action blocks, discounting dialogue-only openings unless accompanied by physical action verbs (slap, strike, fall, explosion) or tight push-ins.

FAILURE MODE 2: CONTEXT-BLIND LORE TRAPS

The Defect: Scenes where characters dispute legal documents, Father Thomas's council petition, or Lord Sterling's couriers (e.g., Episodes 20, 23, 49).

Why It Fails: While dramatic to a binge-viewer who watched 25 episodes, it is completely meaningless jargon to a cold social media user and generates zero emotional investment.

Fix Implemented: Standalone Coherence filter penalizes secondary institutional lore, reserving these clips for mid-season retargeting while barring them from Top-of-Funnel spend.

FAILURE MODE 3: SUB-SECOND ACOUSTIC DECAY CLIPPING

The Defect: Cutting strictly at the screenplay's printed millisecond timestamp clips lingering vocal reverberations, actor inhales, or orchestral swell decays.

Why It Fails: Screenplay timestamps track text delivery, not acoustic room acoustics or emotional pauses.

Fix Implemented: Output timeline manifests automatically inject a **+0.40s audio safety padding** to prevent harsh acoustic truncation in automated video assembly.

FAILURE MODE 4: DISJOINT CROSS-SCENE JUMP CUTS

The Defect: Multi-segment stitching across disparate environments (e.g., raining woods to indoor dining hall) disorients the viewer, feeling like a streaming glitch.

Why It Fails: Algorithmic concatenation ignores visual lighting and costume continuity.

Fix Implemented: Multi-segment engine enforces strict causal continuity rules and flags a mandatory 0.2s transition effect (sound whoosh/white flash) between segments.

5. AI TOOLS USED & HONEST REFLECTION (WHERE AI GOT IT WRONG)

AI Tools Leveraged: Gemini 1.5 Pro and GPT-4o were utilized for rapid screenplay regex parsing development, marketing archetype ontology formulation, and heuristic weighting calibration.

Critical Failure Modes in Brute-Force Prompting: When given raw screenplay text and prompted to *"extract the best 60-second promos"*, generic LLMs consistently exhibited severe flaws:

- **Hallucinated Timestamps & Drift:** LLMs hallucinated non-existent cut points or skipped script lines, yielding clips that varied from 14s to 85s and broke strict duration constraints.
- **Sentence Mid-Utterance Slicing:** Arbitrary semantic cuts severed dialogue mid-syllable, destroying audio and performance flow.
- **Architectural Solution:** We strictly stripped the LLM of timestamp calculation authority. Boundaries and durations are computed 100% deterministically in code, reserving AI heuristics purely for qualitative semantic scoring and virality archetype classification.

6. PROTOTYPE LINK & SUBMISSION DELIVERABLES

PRODUCTION REPOSITORY & RUN COMMANDS

GitHub Repository:
https://github.com/chandraneel07/kuku_promo_finder

- `git clone https://github.com/chandraneel07/kuku_promo_finder.git`
- `npm run pipeline (Parses 50 episodes in 0.75s, outputs 148 ranked promos)`
- `npm start (Runs interactive web studio at http://localhost:3000)`

Deployed live on Vercel with mobile/desktop responsiveness, dataset filtering, and full CSV export capability.

GENERATED RAW OUTPUTS PACK (ALL 50 EPISODES)

- `outputs/promos_ranked.csv`: Complete spreadsheet table with millisecond timestamps, ordered scenes, duration, hooks, cliffhangers, and rationales for all 148 promos.
- `outputs/promos_ranked.json`: Machine-readable payload for automated NLE timeline injection.
- `outputs/quality_check_report.json`: Automated diagnostic audit across all 4 failure modes.
- `outputs/promos_summary.md`: Human-readable executive markdown catalog.